

[Dashboard](#)[my courses](#)[AMCN SS 2022](#)[wireless transmission](#)[Quiz: Wireless Transmission - Modulation and Coding](#)

Started at	Sunday, April 17, 2022 at 1:06 p.m
status	Completed
Finished at	Sunday, April 17, 2022, 1:12 p.m
Spent time	5 minutes 59 seconds
valuation	9.00 out of 9.00 (100 %)
feedback	Congratulations, excellent performance (90-100%).

question **1**

Complete

Score 1.00 out
of 1.00

How can we optimize the throughput of a single transmitter-receiver pair under given bandwidth constraints as well as fixed signal and interference values?

Choose one or more answers:

- ☒ a. Adapt modulation scheme
- ☒ b. Use multiple antennas
- ☒ c. Adapt coding scheme

question **2**

Complete

Score 1.00 out
of 1.00

How is direct communication between mobiles (D2D) a solution to improve the network capacity of cellular networks?

Choose one or more answers:

- ☒ a. Proximity gain due to smaller distances between sender and receiver
- ☒ b. Gain due to one instead of two wireless communications (sender to base station, base station to receiver)

question **3**

Complete

Score 1.00 out
of 1.00

What are the available choices of adding information to a periodic signal?

Choose one or more answers:

- ☒ a. phase modulation
- ☒ b. Frequency modulation
- ☒ c. amplitude modulation
- ☐ i.e. spatial modulation
- ☐ e. bandwidth modulation

Frage **4**

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the advantage of QPSK over 16-QAM?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Lower latency
- ☒ b. Lower error rate
- ☐ c. Higher data rate
- ☒ d. Fewer problems with multipath propagation

Frage **5**

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the benefit of frequency hopping?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Reduced sensitivity to broadband interference
- ☒ b. Reduced sensitivity to narrowband interference and noise
- ☒ c. Distribution of the signal over a wider bandwidth
- ☐ d. Higher spectral efficiency (bits per Hz)



Frage 6

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the benefit of spectrum spreading technology?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Reduced sensitivity to narrowband interference and noise
- ☐ b. Higher spectral efficiency (bits per Hz)
- ☐ c. Reduced sensitivity to broadband interference
- ☒ d. Distribution of the signal over a wider bandwidth

Frage 7

Vollständig

Erreichte
Punkte 1,00 von
1,00

What limits the use of higher order modulation schemes, as e.g. 64-QAM?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. High SINR needed for proper transmission
- ☒ b. Highly sensitive to distortions in the phase of the signal
- ☒ c. Highly sensitive to multipath problems
- ☒ d. Highly sensitive to distortions in the amplitude of the signal

Frage 8

Vollständig

Erreichte
Punkte 1,00 von
1,00

Which modulation schemes apply some sort of amplitude modulation?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. 9-QAM
- ☐ b. QPSK
- ☒ c. 16-QAM
- ☒ d. 64-QAM
- ☐ e. GMSK

Frage 9

Vollständig

Erreichte
Punkte 1,00 von
1,00

Why is amplitude modulation in its pure form rarely used in practice?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Receivers are highly sensitive to changing channel conditions
- ☒ b. Highly sensitive to multipath problems
- ☐ c. More problems with spatial propagation
- ☐ d. Higher pathloss compared to other modulation schemes
- ☐ e. Highly sensitive to frequency changes

Vorhergehende Aktivität

◀ Quiz: Wireless
Transmission - Multiplexing
(Splitting up the Radio
Resource)

Direkt zu:



next activity

Quiz: Wireless Transmission
- Cellular Networks (Capacity
and Interference) ▶



imprint

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